Project Brief: Lylac Jewels Inventory and Order Management System

Project Motivation

For my semester design project, I plan to develop an inventory and order management system for Lylac Jewels. Lylac Jewels is a real jewelry business owned by my sister, and its website is currently in the planning and development process. The business needs an organized way to manage its jewelry products, available quantities, customer orders, and order statuses. Managing this information manually can become difficult as the number of products and customers increases. It can also lead to mistakes, such as accepting an order for an unavailable item or forgetting to update the status of an order.

I chose this project because it solves a real business problem and gives me an opportunity to create software that could be used outside the classroom. My sister can act as the client by explaining the needs of the business and providing feedback throughout the project. Her feedback will help me understand how requirements can change during software development. This project will also give me practical experience that I can include in my portfolio and resume.

System Overview

The proposed system will be a web application that helps the owner of Lylac Jewels manage products, inventory, and customer orders. The system will include a secure administrative area where the owner can sign in and access a dashboard. From the dashboard, the owner will be able to add new jewelry products, update existing product information, enter prices, organize products by category, and change available quantities.

The order-management section will allow the owner to record and review customer orders. Each order will contain information about the customer, the products ordered, the quantity, and the current order status. The owner will be able to change the status to pending, confirmed, shipped, delivered, or canceled. The system should automatically reduce the available inventory when an order is confirmed. If time allows, I would also like to include product-image uploads, low-stock notifications, search and filtering, and basic sales reports. Complicated online payment processing will not be part of the initial project scope.

Proposed Technologies

I plan to use Python and Flask for the back end of the application. Python is appropriate because I already have some experience using it, and Flask is a useful framework for developing a web application without adding unnecessary complexity. I plan to use HTML and CSS to create the pages and design the user interface. I may also use JavaScript for features that require interaction without reloading the entire page.

SQLite will be used during the early development stage because it is simple to configure and works well for a small application. If the system requires a larger database later, it could be moved to MySQL. The database will contain related tables for products, categories, customers, orders, and order items. I will use GitHub for version control so I can save the project's history, manage changes, and show my progress through regular commits.

Target Audience and Users

The main user of the system will be the owner of Lylac Jewels. She will interact with the system through a secure web dashboard. The dashboard should be simple enough for someone without advanced technical knowledge to use. It will provide forms for entering product information, updating stock, and managing orders.

Customers may also interact with a public part of the system by browsing available jewelry and viewing product information. A customer-facing ordering feature may be added if the main administrative features are completed

within the available time. The most important goal is to make the business-management section functional and easy to use.

Project Structure and Responsibilities

This will be an individual project. I will be responsible for gathering requirements, planning the system, designing the database, creating the user interface, writing the back-end code, testing the features, fixing errors, and documenting the project. I will also create and maintain the GitHub repository and make regular commits during development.

I plan to divide the work into smaller stages. I will first identify the system requirements and design the database. After that, I will create the basic pages and administrative login. I will then develop the product and inventory features, followed by the order-management features. The final stages will include testing, improving the interface, and preparing the system for demonstration.

Personal Programming Contribution

One major feature I will personally implement is the product and inventory management function. This feature will allow the owner to add a product, enter its name, category, description, price, and available quantity, and later update or remove that product. I will also program the connection between confirmed orders and inventory so that the product quantity is reduced when an order is confirmed. The system should prevent an order from being confirmed when there is not enough inventory available.

GitHub repository URL: https://github.com/JashJivani23/LYLAC-JEWELS